



To Catch and Reverse a Quantum Jump Mid-flight



**Innovators
Under 35**

Zlatko Minev, Ph.D.

IBM Quantum Research, TJ Watson

@zlatko_minev

Quantum

Discreteness

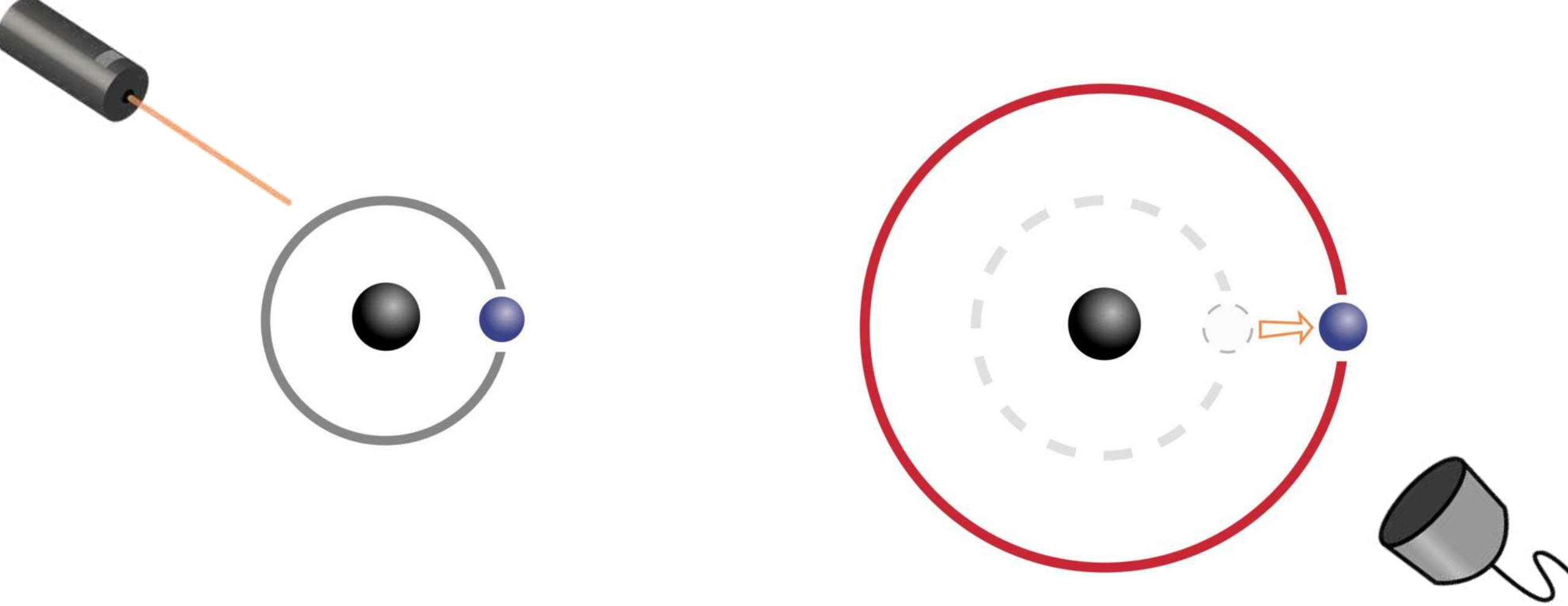


Unpredictability



Predict?





$|0\rangle$



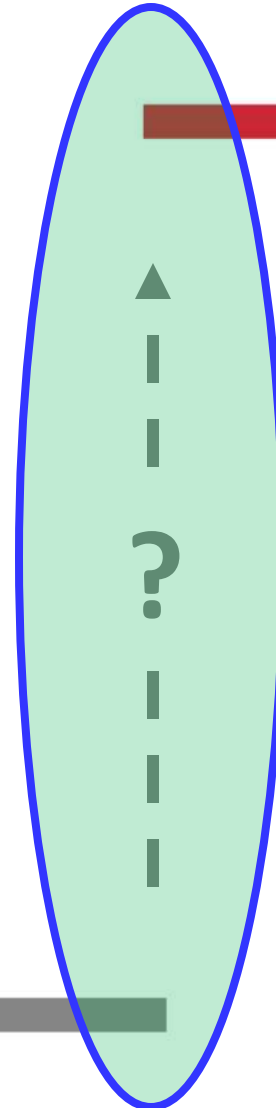
$|1\rangle$



Quantum Jump: Until Now



$|1\rangle$

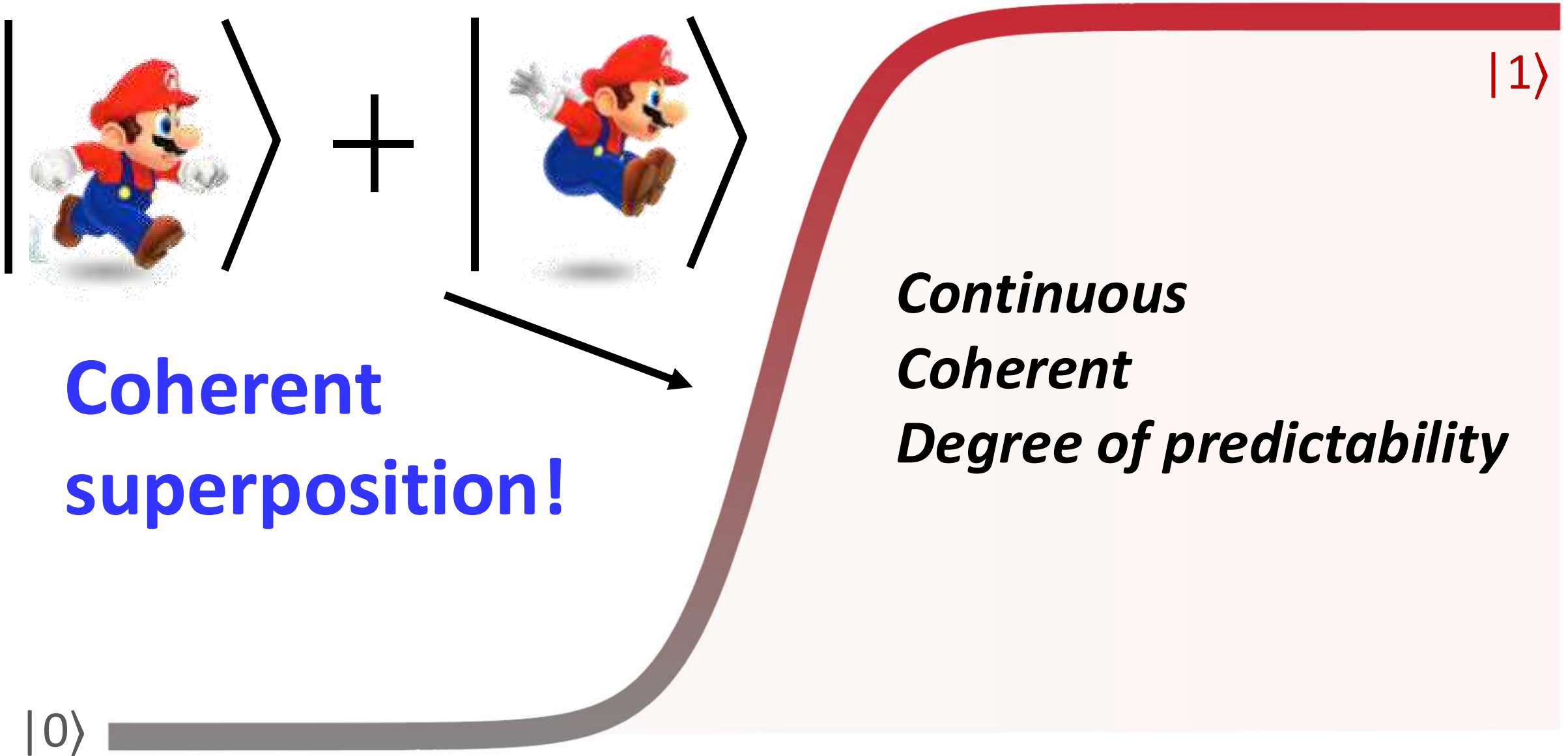


Discrete?
Instantaneous?
Unpredictable?

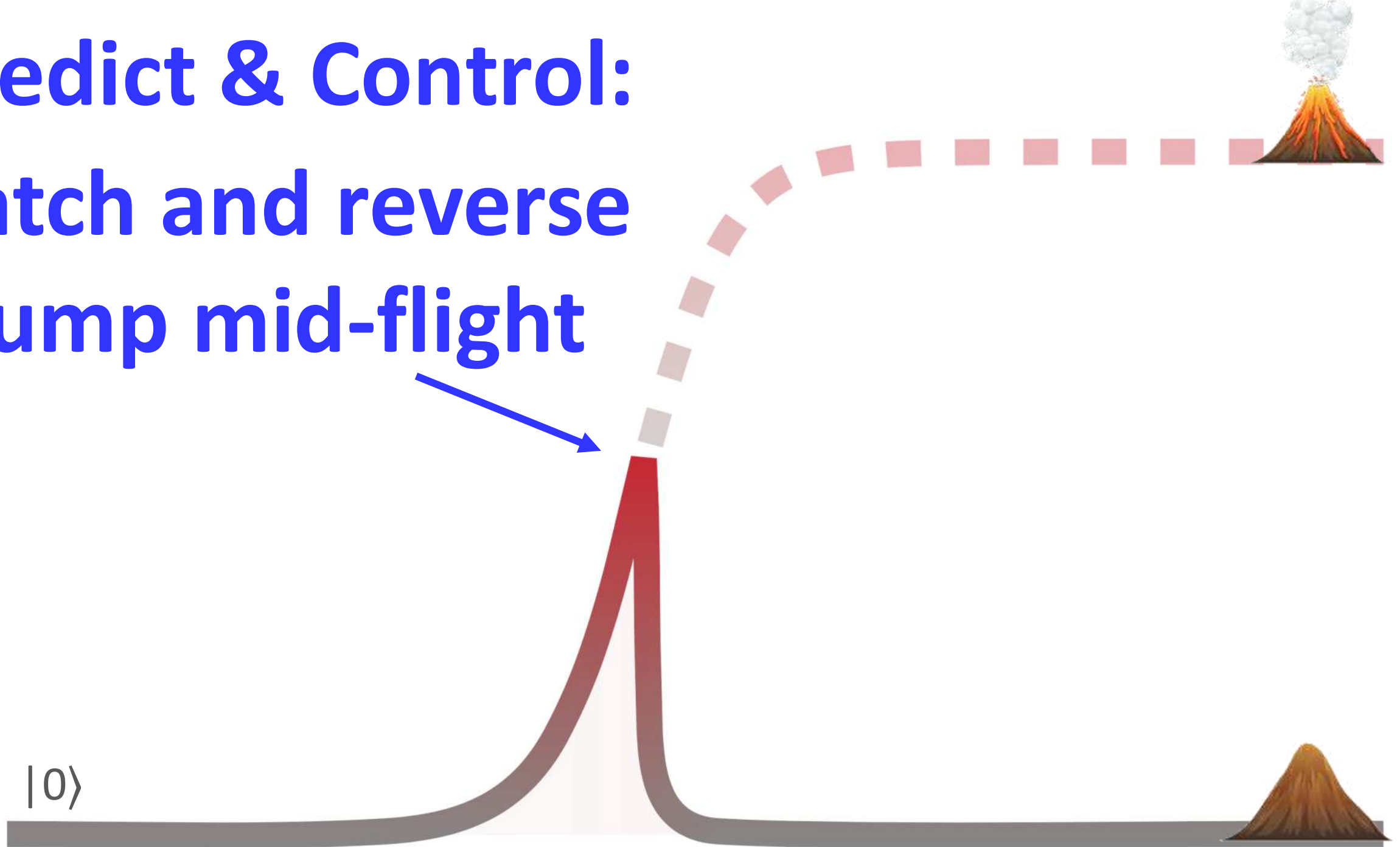
$|0\rangle$



The Real Quantum Jump



Predict & Control: Catch and reverse jump mid-flight



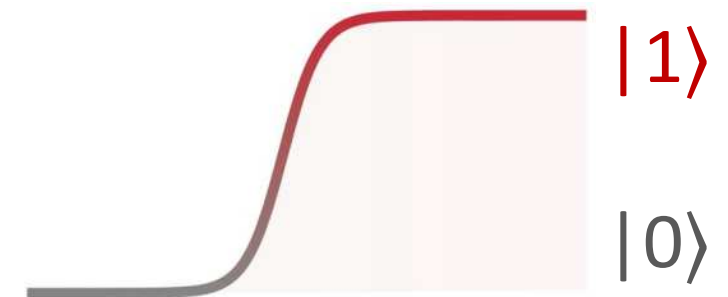
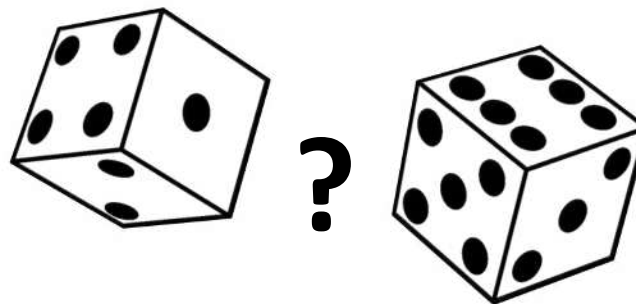
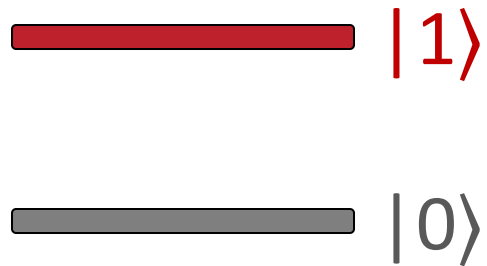
Bohr



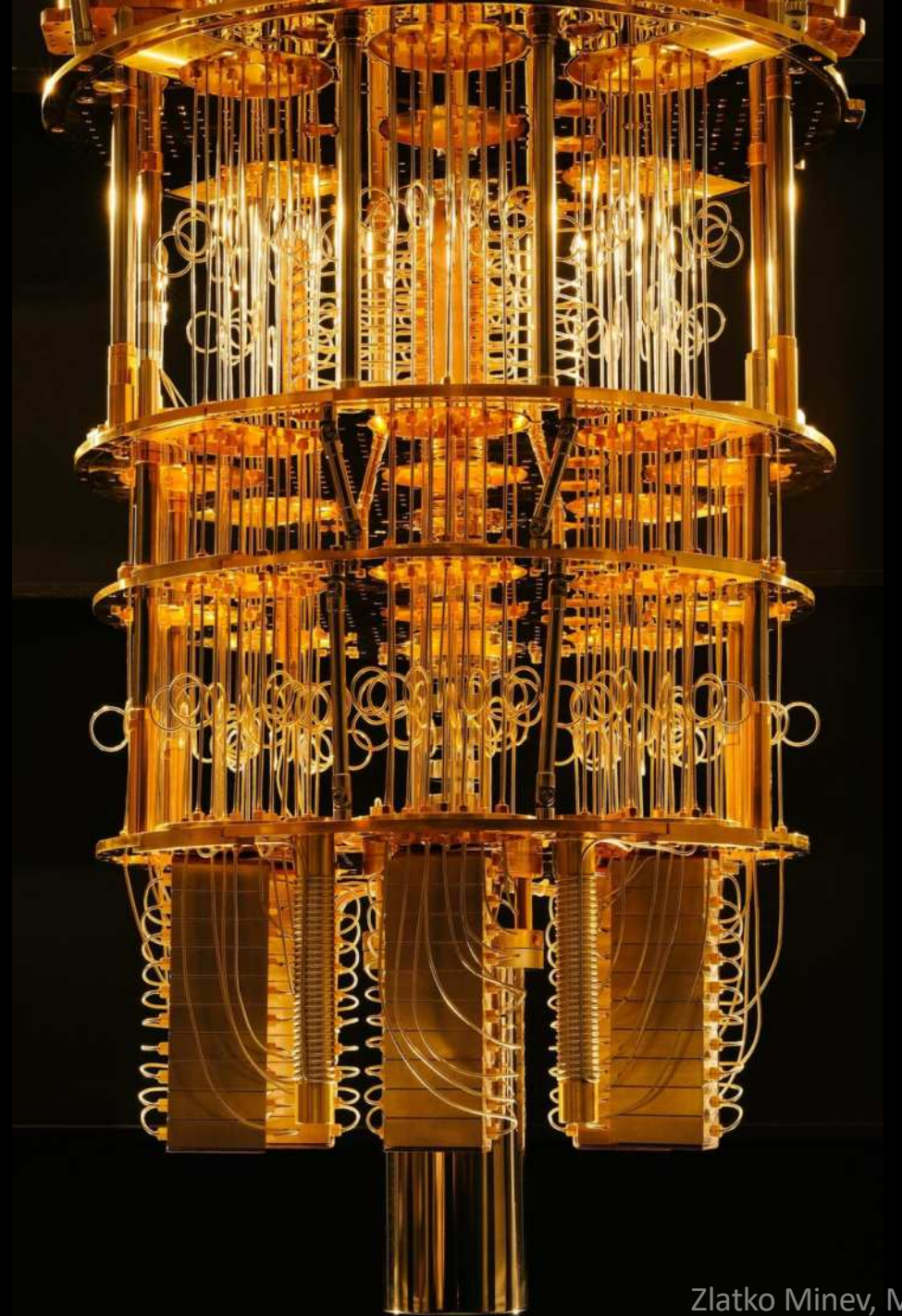
Einstein



Schrödinger



Quantum Computing: Error correction Quantum Sensing



There is more to the story of quantum physics ...

Not quite as discrete and random as we thought ...

Opens doors to otherwise inaccessible technology ...



Thank you!



**Innovators
Under 35**

Zlatko Minev, Ph.D.

IBM Quantum Research, TJ Watson

@zlatko_minev